Bipin Kumar

Data Science Engineer | AI/ML Developer | Python & Deep Learning
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Professional Summary

Data Science and AI/ML Engineer with hands-on experience building end-to-end machine learning pipelines, LLM-powered applications, and data-driven web products. Proficient in Python, Scikit-learn, TensorFlow, LangChain, and the MERN stack. Passionate about solving real-world problems through intelligent systems. Seeking Data Science / ML Engineer roles.

Education

Jagannath University Jaipur, India
Bachelor of Technology (B.Tech) in Computer Science Jul 2023 – Expected Jul 2027
CGPA: 7.4 | *Coursework:* Machine Learning, Data Structures & Algorithms, DBMS, Statistics, Computer Networks, OOP

Technical Skills

- **Languages:** Python, SQL, JavaScript (ES6+), R (basics)
- **ML/DL:** Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, LightGBM, ONNX, HuggingFace Transformers
- **Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, Excel
- **NLP / LLM:** LangChain, LlamaIndex, RAG Pipelines, OpenAI API, Groq, NLTK, SpaCy, Sentence Transformers
- **Full-Stack (MERN):** React.js, Node.js, Express.js, MongoDB, REST APIs, Mongoose, Tailwind CSS, Redux
- **DevOps & Tools:** FastAPI, Streamlit, Docker, Git, ChromaDB, FAISS, MongoDB Atlas, PostgreSQL, Jupyter, VS Code

Experience

CodeWithHarry Remote
Data Science Trainee Sep 2025 | Certificate
- Completed **The Ultimate Job Ready Data Science Course** covering the full ML pipeline: EDA, feature engineering, model training, evaluation, and deployment using Python, Pandas, Scikit-learn, and Matplotlib.
- Applied statistical analysis and supervised/unsupervised ML techniques on real-world datasets, achieving production-ready model workflows.

Udemy Remote
Python Programming Trainee Sep 2025 | Certificate
- Completed **Complete 2025 Python Bootcamp** (15 hrs) by Haris Ali Khan, mastering Python fundamentals, OOP, file I/O, and libraries essential for data science and automation workflows.
- Built multiple mini-projects using Python covering data manipulation, API integration, and scripting for ML preprocessing tasks.

Projects

RAG-Powered Document Q&A System | *Python, LangChain, ChromaDB, OpenAI API, FastAPI, React, MongoDB*
- Built a **Retrieval-Augmented Generation (RAG)** pipeline ingesting PDFs and docs via LangChain, embedding with **Sentence Transformers**, and storing vectors in **ChromaDB** for semantic search with **<300ms** query latency.
- Developed a React + FastAPI full-stack interface with MongoDB session storage, JWT auth, and Groq LLM integration for context-aware, hallucination-reduced answers across multi-document corpora.

Movie Recommendation System | *Python, Scikit-learn, Pandas, Node.js, Express, React, MongoDB*
- Engineered a **hybrid recommendation engine** combining **content-based filtering** (TF-IDF + cosine similarity) and **collaborative filtering** (SVD matrix factorization) on the MovieLens 1M dataset, achieving **RMSE 0.87**.
- Built a MERN stack web application with **MongoDB Atlas** for user preference storage, RESTful API via Express, and a React UI with real-time search and personalized movie cards.

Customer Churn Prediction Dashboard | *Python, XGBoost, SHAP, FastAPI, React, MongoDB, Streamlit*
- Trained an **XGBoost classifier** on telecom churn data with SMOTE oversampling, feature selection via **SHAP**, and hyperparameter tuning (GridSearchCV), achieving **92% AUC-ROC**.
- Deployed via **FastAPI** backend with MongoDB logging and built a dual interface: Streamlit analytics dashboard for data teams and a React admin panel with churn risk scoring per customer.

Real-Time Sentiment Analysis Pipeline | *Python, HuggingFace, BERT, Node.js, Express, React, MongoDB*
- Fine-tuned **BERT (bert-base-uncased)** on the IMDb and Twitter datasets for multi-class sentiment classification, achieving **91% F1-score**; exported to ONNX for **3× faster inference**.
- Integrated into a MERN stack app where users submit reviews via React UI, Express routes calls to the Python model server, and MongoDB stores predictions with timestamped sentiment history charts.